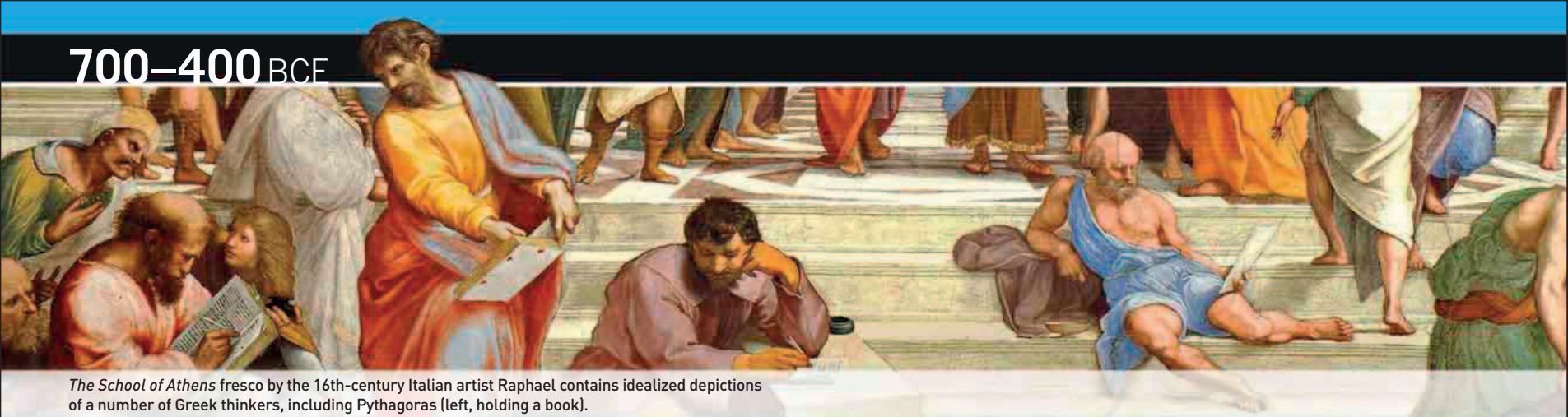




The School of Athens fresco by the 16th-century Italian artist Raphael contains idealized depictions of a number of Greek thinkers, including Pythagoras (left, holding a book).

AS EARLY AS 2300 BCE, the Babylonians had developed a sexagesimal number system (based on **writing numbers** in multiples of 60) and the principle of position (where numbers in different positions represent different orders of magnitude). By 700 BCE they sometimes used a **marker to indicate a null value (zero)**.

The screw pump (or Archimedes Screw) is a cylindrical pump with a central shaft surrounded by inner blades in the shape of a spiral and encased in wood. As the shaft is rotated, **water is pulled up the spiral**, transferring it from a lower to a higher level. The invention of the pump is traditionally ascribed to the ancient Greek mathematician Archimedes (287–212 BCE) in

around 250 BCE, but it was probably first invented much earlier, in the 7th century BCE under the rule of King Sennacherib of Assyria to water his palace gardens at Nineveh.

By the 1st millennium BCE, the Babylonians had begun to make maps of larger areas. By around 600 BCE they had produced a **“world map,”** which showed the **city of Babylon** in relation to eight surrounding regions. The first known **Chinese map**, found on an engraved bronze plaque in the tomb of King Cu of Zhongshan, was a plan of the king’s proposed necropolis.

The ancient Greek cartographic tradition began in Ionia in the 6th century BCE. **Anaximander** (c.611–546 BCE) is said to have drawn the **first world map** that

showed Earth surrounded by a great ocean. Hecataeus of Miletus (c.550–480 BCE) also drew a map to accompany his *Survey of the World* that showed three great continents, Libya (Africa), Asia, and Europe.

The first evidence of scientific (as opposed to supernatural) thinking about the nature of the world came from **ancient Greek philosophers** in the 6th and 5th centuries BCE. Thales of Miletus (b. c.620 BCE) believed that **water was the fundamental material of the universe**, and that earthquakes happened when the surface of Earth rocked on the watery surface on which it floated. In contrast, **Anaximander**, who was also from Miletus, believed that the prime material of the universe was apeiron, a substance that preceded air, fire, and water. He also put forward an **early evolutionary theory**, suggesting that humans had developed from a type of fish.

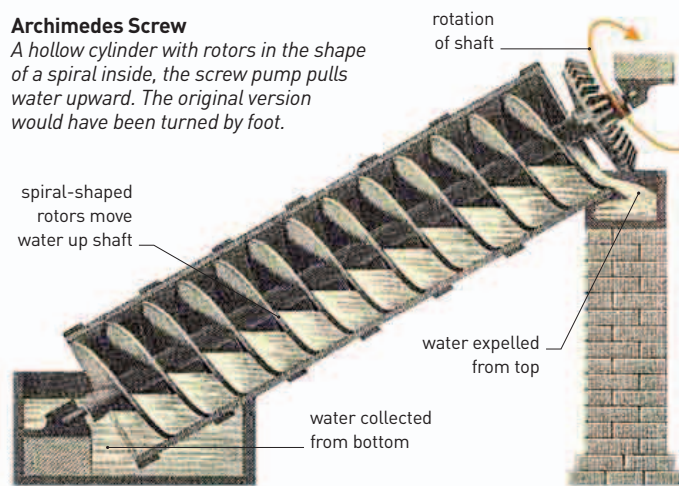
The **first atomic theory** was also proposed by a Greek, the philosopher Democritus of



Ancient map
This Babylonian map from around 600 BCE shows the relationship between Babylon and other important places in West Asia, including Assyria and Urartu.

Archimedes Screw

A hollow cylinder with rotors in the shape of a spiral inside, the screw pump pulls water upward. The original version would have been turned by foot.



700 BCE Babylonians use a marker to represent a null value (zero)

c.700 BCE Cylindrical pump, later known as Archimedes Screw, is in use by Assyrians to pump water

600 BCE Babylonians produce a “world map” showing the city of Babylon and eight surrounding regions

c.580 BCE Thales of Miletus puts forward the idea that water is the basic material of the universe

c.550 BCE Anaximander puts forward an early theory of evolution

c.530 BCE Eupalinos builds a water tunnel through a hillside on Samos

c.530 BCE Pythagoras develops the theorem about the ratios of the sides of a right-angled triangle, now known as Pythagoras’s theorem